HUGO'S APPLIANCE DIAGNOSTIC SYSTEM

APPLIANCE DIAGNOSTIC DATA RECORD

Page 1 - Context, safety baseline, symptom classification, and diagnostic path

CASE ID: _____
TECHNICIAN: Hugo Liu
DATE: _____

0. SOURCE REFERENCE & VISIT TRACEABILITY

ISO/RCA TRACEABILITY

SCHEMATIC / TECH SHEET REF. #	SERVICE MANUAL / PUB #	CUSTOMER / SITE REF.	P.O. #
Loc: Kickplate / Console / Back			

Visit: ☐ 1st Visit ☐ Recall Prev. Case ID: _____

Account: ☐ Retail / COD ☐ Property / Landlord ☐ 3rd-Party Warranty ☐ OEM / Factory

1. EQUIPMENT & SYMPTOM CLASSIFICATION

APPLIANCE TYPE	BRAND	MODEL NO.	SERIAL NO.
☐ REF ☐ DW ☐ DRY ☐ WSH ☐ RGE-OVN ☐ OTHER			

APPROX. AGE _____ | ERROR CODE / DISPLAY MESSAGE **Code:** _____ | ☐ Active ☐ Stored History

REPORTED SYMPTOM / CUSTOMER COMPLAINT

PRIMARY SYMPTOM - SELECT ONE (ROOT DECISION BRANCH)

○ No Power ○ No Start ○ No Heat / Cool ○ Leak / Drain ○ Noise / Vibration ○ Stops Mid-Cycle ○ Intermittent

SECONDARY SYMPTOM(S) - SELECT ALL THAT APPLY

☐ Noise ☐ Odor ☐ Leak ☐ Slow / Weak ☐ Overheat ☐ Trips Breaker ☐ Other

FAILURE TIMING / TRIGGER CONDITION

○ Immediate at Start ○ During Cycle (Min: _____) ○ Specific Phase: _____

2. INITIAL SAFETY, ISOLATION & UTILITY BASELINE

SAFETY / ACCESS CHECKS

☐ Electrical Isolation Verified ☐ LOTO if Applicable ☐ Power / Breaker Checked ☐ Cord / Plug Inspected

☐ Water / Drain / Vent ☐ Leak / Moisture Checked ☐ Burn / Odor Checked ☐ Mechanical / Access OK

UTILITY / ENVIRONMENT BASELINE

Supply Type: ☐ 120V 1Ph ☐ 120/240V Split-Ph

L1-N: _____ V **L2-N:** _____ V **L1-L2:** _____ V

Water Temp: _____ deg F **Pressure:** Static: _____ psi | Dyn: _____ psi

Gas: ☐ NG ☐ LP **Ambient / Notes:** _____

3. DIAGNOSTIC SEQUENCE STRUCTURED DATA FOR FUTURE DECISION TREE

HYPOTHESIS DRIVEN TESTING FLOW

STEP	HYPOTHESIS / POSSIBLE CAUSE	TEST / MEASUREMENT Component / Terminal Pin / Mode State: Static (Unplugged) / Live (Powered)	EXPECTED (V/Ohm/A/Hz/deg F/psi)	ACTUAL Measured Value + Unit	RESULT P / F	DECISION / NEXT STEP -> Step N / Confirmed / Clear
1						
2						
3						
4						
5						
6						

HUGO'S APPLIANCE DIAGNOSTIC SYSTEM | PAGE 2: CLOSURE & RULE EXTRACTION

CASE ID: _____ **MODEL NO.:** _____ **DATE:** _____

4. ROOT CAUSE, FAILURE MODE, CONFIDENCE & REPAIR ACTION

CONFIRMED ROOT CAUSE

CONFIDENCE
○ High ○ Medium ○ Low
*If Med/Low: note missing condition or risk factor below

FAILURE MODE - SELECT PRIMARY MECHANISM
☐ Open ☐ Short ☐ Stuck ☐ Restricted ☐ Thermal ☐ Loose/Corroded ☐ Logic/Firmware ☐ Other

REPAIR ACTION / CORRECTIVE WORK PERFORMED

PARTS REQUIRED / INSTALLED

OEM PART # / SKU

PARTS / SUPPLY STATUS
☐ In Stock ☐ Ordered ☐ Backorder ☐ Customer-Supplied ☐ No Part Required

5. POST-REPAIR VERIFICATION & FINAL STATUS

VERIFICATION READINGS / TEST CYCLE / SAFETY CHECK
☐ Current / Amps in spec ☐ Relevant Cycle Completed ☐ No Leak Verified ☐ Ground Safety Checked

FINAL STATUS
☐ Operating Normally ☐ Further Testing ☐ Parts Required ☐ No Fault Found ☐ Repair Declined ☐ Not Repairable

FIRST-TIME FIX / RECALL OUTCOME
○ First-Time Fix ○ Return Visit Required ○ Recall Resolved

6. OUTCOME CODE, FAILURE CATEGORY & DATA TAGS

CASE OUTCOME CODE
☐ RP ☐ CA ☐ WI ☐ NTF
☐ BER ☐ PR ☐ DECLINED

FAILURE CATEGORY / SUBSYSTEM
☐ Pwr ☐ Ctrl ☐ Sens ☐ Mtr ☐ Pmp
☐ Heat ☐ Refrig ☐ Wtr/Drn ☐ Mech ☐ Wire

DATA TAGS / KEYWORDS (FUTURE DATABASE INDEXING)

7. TECHNICIAN SUMMARY / LESSONS LEARNED

TECHNICAL SUMMARY / OBSERVATIONS / REPEATABLE PATTERN

8. RULE EXTRACTION / DIAGNOSTIC TREE MAPPING

EXPERT SYSTEM KNOWLEDGE BASE

Translate the completed field record into a reusable rule-engine / DAG node after the case is closed.

ROOT / SYMPTOM ID	**CONTEXT CONDITION(S)**
NODE ID	**TEST / METRIC**
BRANCH CONDITION	**TRUE -> NEXT NODE / LEAF**
FALSE -> ALTERNATIVE NODE	**ROOT CAUSE ID / OUTCOME CODE**

PRODUCTION RULE / IF-THEN EXTRACTION

IF _____
(Root Symptom / Context)

AND _____
(Gate Test / Measurement)

THEN _____
(Confirmed Root Cause / Part / Action)

Data principle: Preserve raw measurements, test state, context, and branch outcome before conclusion; reusable across a rule engine / DAG.